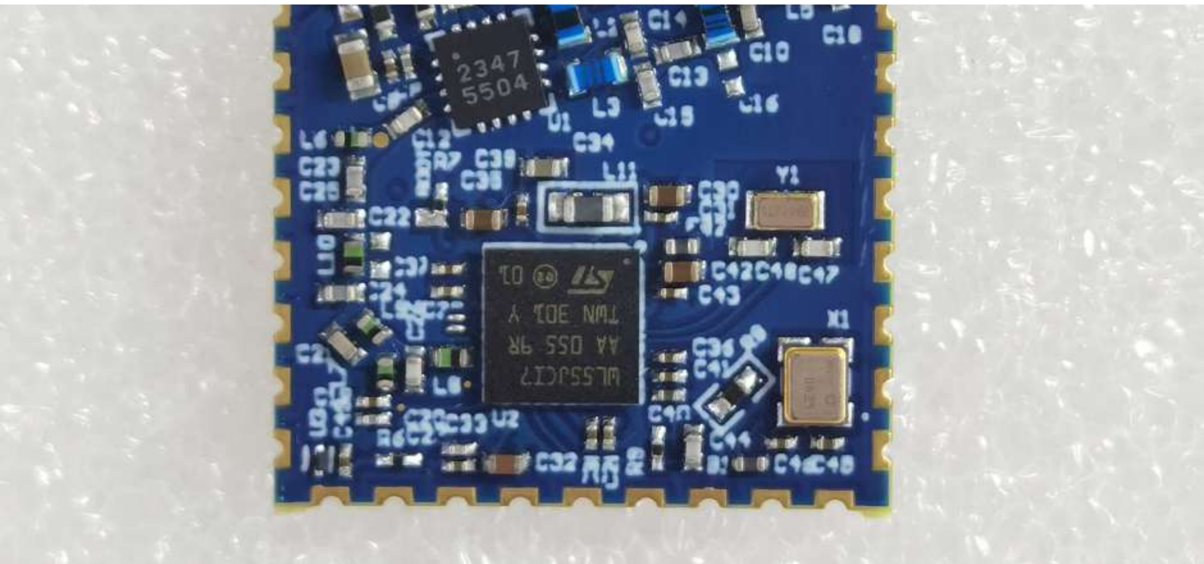


# SMD Module for Argos Communication

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The Argos SMD module is a small PCB (2×2 cm) designed to function as either a standalone unit or a coprocessor for any new design requiring satellite communication (uplink only, no downlink).

The module is based on the STM32WL55 processor, combined with the GR5504 power amplifier, allowing communication at up to 26 dBm. This module, which has passed CLS certification, is ready for use.

All GitHub repositories mentioned in this document are currently private.

https://github.com/arribada/argos-smd-hw

# 1. Size and Dimensions

- Size: 20×20mm
- Height: 2mm (without RF shield)
- Weight: <1g
- Altium Integrated library

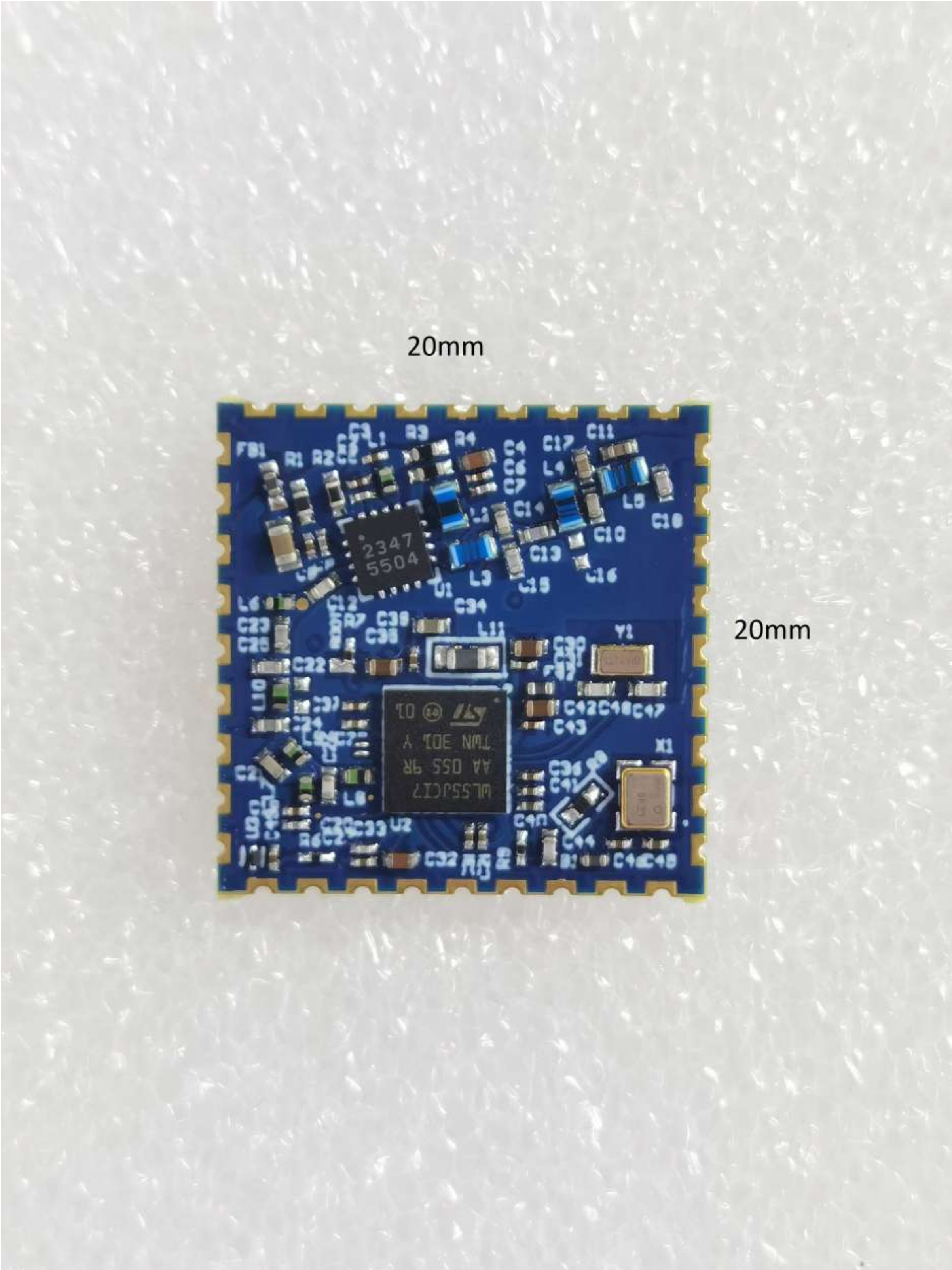
  Arribada-Argos-SMD.LibPkg

  Arribada-Argos-SMD.PcbLib

  Arribada-Argos-SMD.SchLib

- 3D models

  module-BGA_RG-4layers.step




It’s possible to add an RF shield on the board dedicated to integrated the SMD module with the following dimensions:

- Size: 25×25mm

- Height: 3.2mm

# 2. Integration

The module integration requires a power supply for the STM32 and a second one for the external power amplifier. A PI-filter should be added to the antenna output before connecting it to the antenna connector. The TPS63901 is recommended for powering the external PA.

A breakout board is available to test the module and provide a design integration example. This breakout board can be used with a Feather board (from Adafruit) to connect the ARGOS-SMD module. It exposes the ARGOS-SMD module pins to the Feather board and includes a dedicated LDO to power the RF amplifier. You can find resources at the following link.

```
https://github.com/arribada/featherwings-argos-smd-hw
```

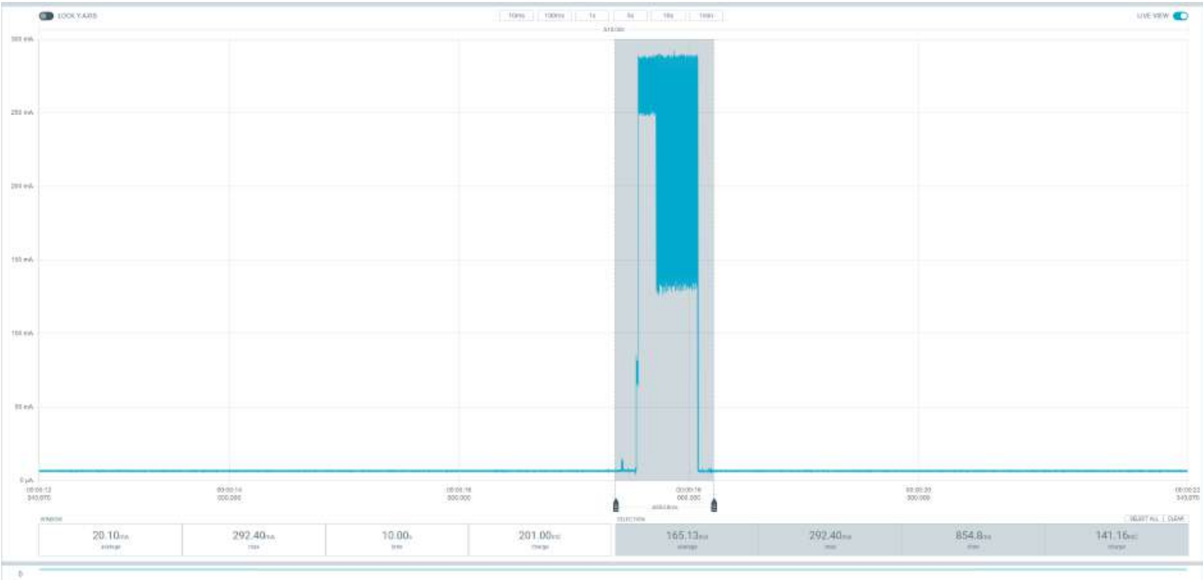
# 3 - Power Supply Voltages

- **VDD (Core Supply Voltage):**
  - **1.8V to 3.6V** operating range.
  - Typical operation at **3.3V**.
- **VBAT (Battery Backup Supply):**
  - Backup battery voltage range from **1.55V to 3.6V**.
  - This is used to power the RTC (Real-Time Clock), backup registers, and some other low-power functionalities when the main supply is off.
- **VDDPA (Power Amplifier Supply):**
  - For the internal power amplifier (PA) dedicated to the radio subsystem.
  - **1.8V to 3.6V** supply voltage.
- **VDD_EXTPA (Power External Amplifier):**

- Typical operation at **3.3V**.

Argos communication power consumption:

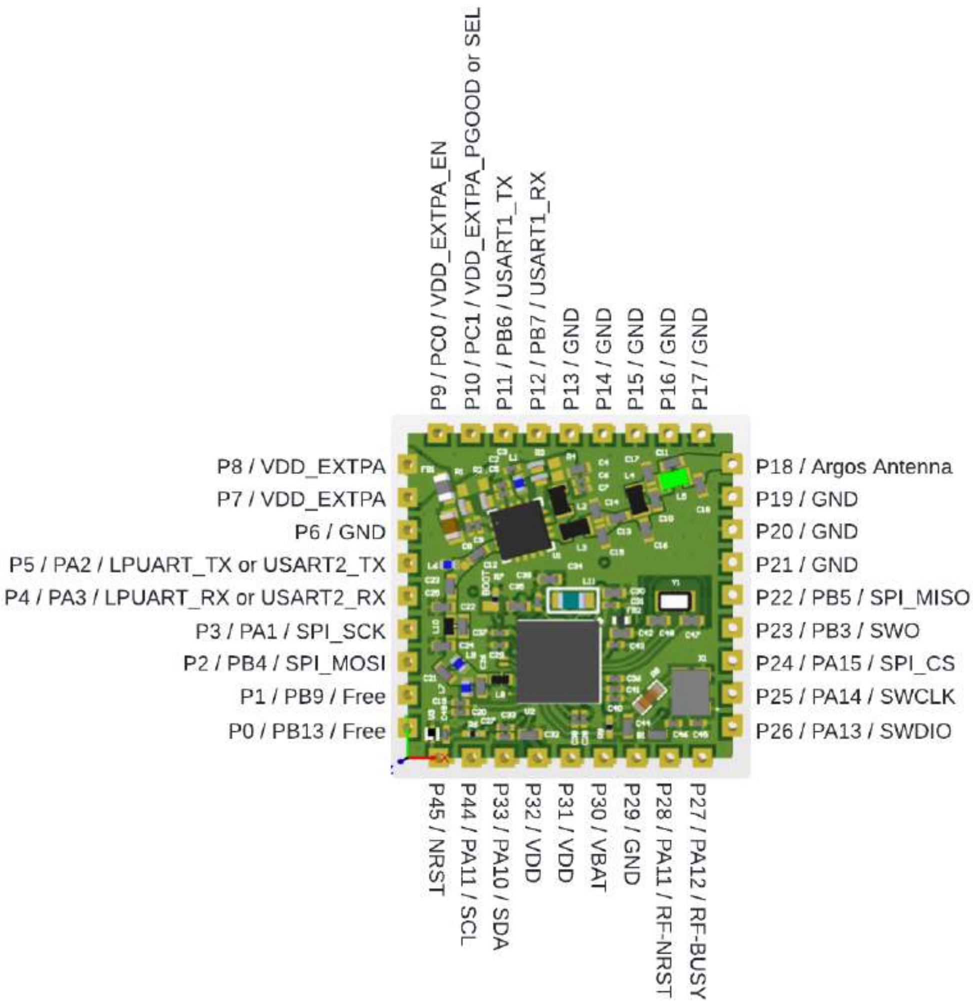
- TX power: 25.8dBm (380mW)
- VPA: 3.40V
- Power max: 292.40mA
- Power mean: 165.13mA



# 4. Pinout

The module features the following pinout configuration:

While it is possible to adapt the pinout definition through programming of the STM32 module, the following pinout has been selected.




System:

- P35 ⇒ nRST
- P31 ⇒ VDD
- P32 ⇒ VDD
- P30 ⇒ VBAT

Power amplifier:

- P7: VDD power amplifier

- P8: VDD power amplifier
- P9: LDO Enable pin
- P10: LDO control pin (GOOD or SEL)
- P18: Antenna out

Debug:

- P26: SWDIO
- P25: SWCLK
- P23: SWDIO
- P27: RF-BUSY
- P28: RF-NRST

I2C:

- P33: SDA
- P34: SCL

SPI:

- P24: NSS
- P3: SPI_SCK
- P2: SPI_MISO
- P22: SPI_MOSI

LPUART:

- P4: UART_RX
- P5: UART_TX

UART:

- P11: UART_RX
- P12: UART_TX

Free:

- p0: PB13 (free)
- p1: PB9 (free)

# 4. Software

The firmware used for the module is based on the latest CLS firmware v6.2.1 and is available on the following GitHub repository.

https://github.com/arribada/SMD-AT-Kineis

The Kineis stack requires credentials (ID, address, secret key) to run properly.

The easiest way to test the module is to use the latest KimGUI interface provided by CLS. This computer application allows communication with the module via UART.

A Zephyr driver can also be used to communicate with the SMD module. For example, when combining the breakout board with an Adafruit Feather board, the following Zephyr driver can be used to test the module.

```
https://github.com/arribada/argos-smd-driver-zephyr
```

It is also possible to use the STM32 example for AT commands to control the Sub-GHz module at this link:

https://github.com/arribada/SMD_AT_Slave

## 4.1. Communication Protocol

The SMD module is configured to communicate via LPUART1 with the following parameters:

- Baud Rate: 9600 bps
- Data Bits: 8
- Stop Bits: 1
- Parity: None
- Flow Control: None

## 4.2. Command Structure

An ID provides by CLS should be configured on the SMD module to allows communication (28 or 32bits ID).

The SMD module use the same KimGUI AT command format, like this :

```
AT+[COMMAND]=[PARAMETER]
```

Example commands:

- "AT+PING=?\r\n to get information
- "AT+TX=<user_data>\r\n to set information

List of commands:

```
        { "AT+VERSION",      10, bMGR_AT_CMD_VERSION_cmd},
        { "AT+PING",          7, bMGR_AT_CMD_PING_cmd},
        { "AT+FW",            5, bMGR_AT_CMD_FW_cmd},
        { "AT+ADDR",          7, bMGR_AT_CMD_ADDR_cmd},
        { "AT+ID",            5, bMGR_AT_CMD_ID_cmd},
        { "AT+SN",            5, bMGR_AT_CMD_SN_cmd},
        { "AT+RCONF",         8, bMGR_AT_CMD_RCONF_cmd},
        { "AT+SAVE_RCONF",   13, bMGR_AT_CMD_SAVE_RCONF_cmd},
        { "AT+LPM",           6, bMGR_AT_CMD_LPM_cmd},
        { "AT+TX",            5, bMGR_AT_CMD_TX_cmd},
        { "AT+CW",            5, bMGR_AT_CMD_CW_cmd},
        { "AT+PREPASS_EN",   13, bMGR_AT_CMD_PREPASS_EN_cmd},
        { "AT+UDATE",         8, bMGR_AT_CMD_UDATE_cmd},
        { "AT+ATXRP",         8, bMGR_AT_CMD_ATXRP_cmd},
```

# 5. Environmental Specifications

The hardware components used in this design have been selected to meet rigorous environmental requirements, ensuring reliable operation in a wide range of conditions. The key specifications are as follows:

1. **Operating Temperature Range:**

- The components are capable of operating in temperatures ranging from **40°C to +85°C**.

2. **Storage Temperature:**
    - The hardware can be safely stored in environments with temperatures between **55°C and +150°C**.
3. **Humidity Resistance:**
    - Operation in environments with **up to 85% to 95% relative humidity**
4. **Moisture Sensitivity:**
    - All components are **Moisture Sensitivity Level (MSL 1)**.
5. **RoHS Compliance:**
    - All components are **RoHS compliant**.

This hardware design leverages industry-standard components, known for their reliability and performance in demanding environments, making it ideal for robust applications such as satellite communication systems.